

Ensemble Methods

- In **simple English**, Ensemble is nothing but **group of items**.
- **Ensemble methods** is a machine learning technique that **combines several base models** in order to **produce one optimal predictive model**.
- It is a powerful method to **improve the performance of the model**.
- The main Aim is to combine all these models , to **increase the Accuracy**.

Contd

- **For Eg:** When you want to buy a laptop, you simply walk up to the store and pick any laptop.



you simply walk up to the

- Generally, You would likely browse a few web portals where people have posted their reviews and compare different laptop models, checking for their features, specifications and prices. You will also probably ask your friends and colleagues for their opinion.

Contd.

- In short, you **wouldn't directly reach a conclusion**, but will **instead make a decision considering the opinions of other people as well**. Here also we can follow the **Ensemble Learning**.
- *An ensemble is itself a supervised learning algorithm*, because it can be **trained and then used to make predictions**.
- **By using the Ensemble Learning, we have 2 objectives:**
 - **1. Reducing the Error**
 - **2. Maintaining the Generalizations.**

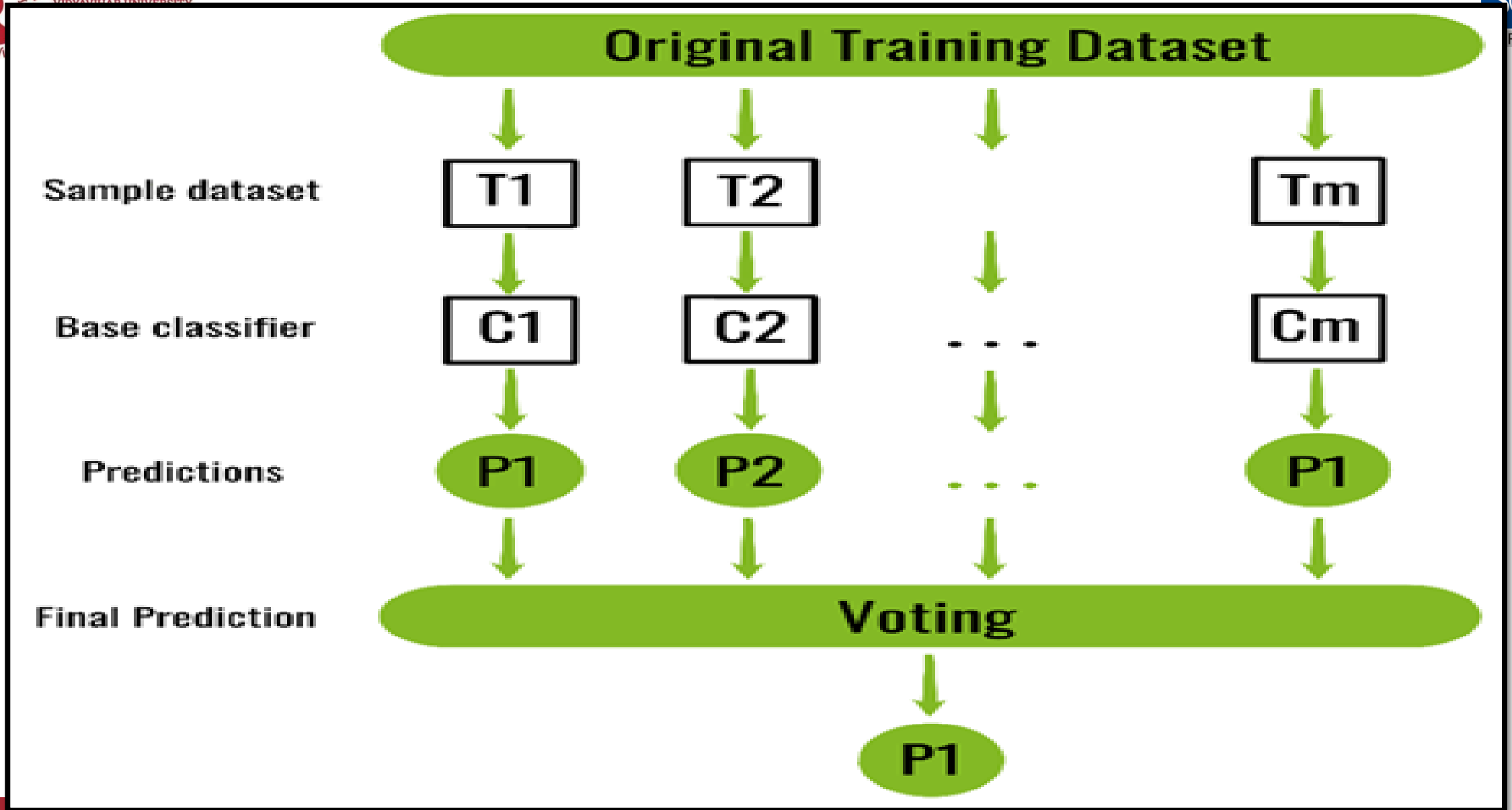
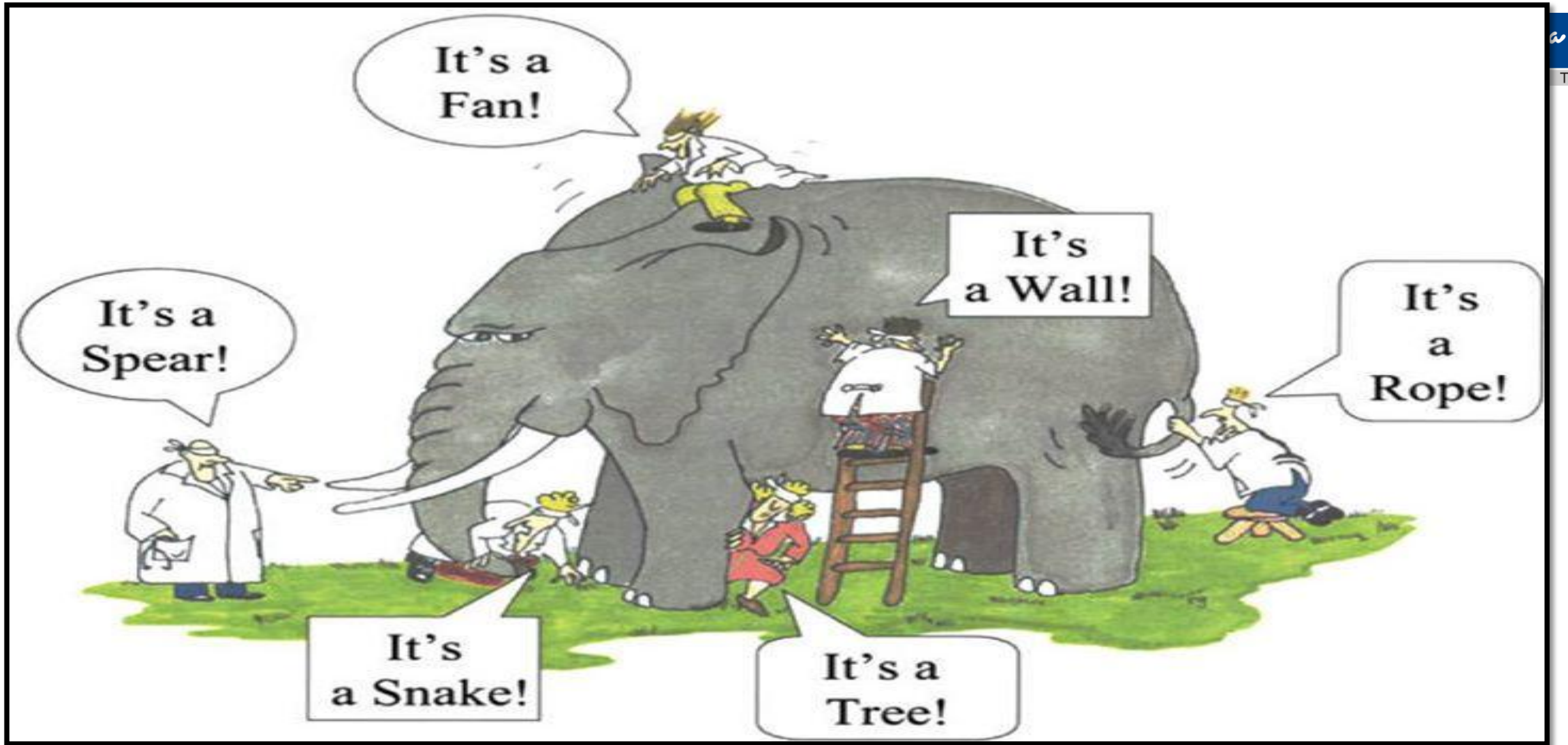


Figure: Ensemble Methods

Example

- Consider the fable of the **blind men** and the **elephant depicted** in the image below.
- The blind men are each describing an elephant from their **own point of view**.
- Their descriptions are **all correct but incomplete**. Their understanding of the elephant would be **more accurate and realistic** if they came together to discuss and combined their descriptions.



The main principle of ensemble methods is to **combine weak and strong learners to form strong learners.**

Types of Ensemble Method

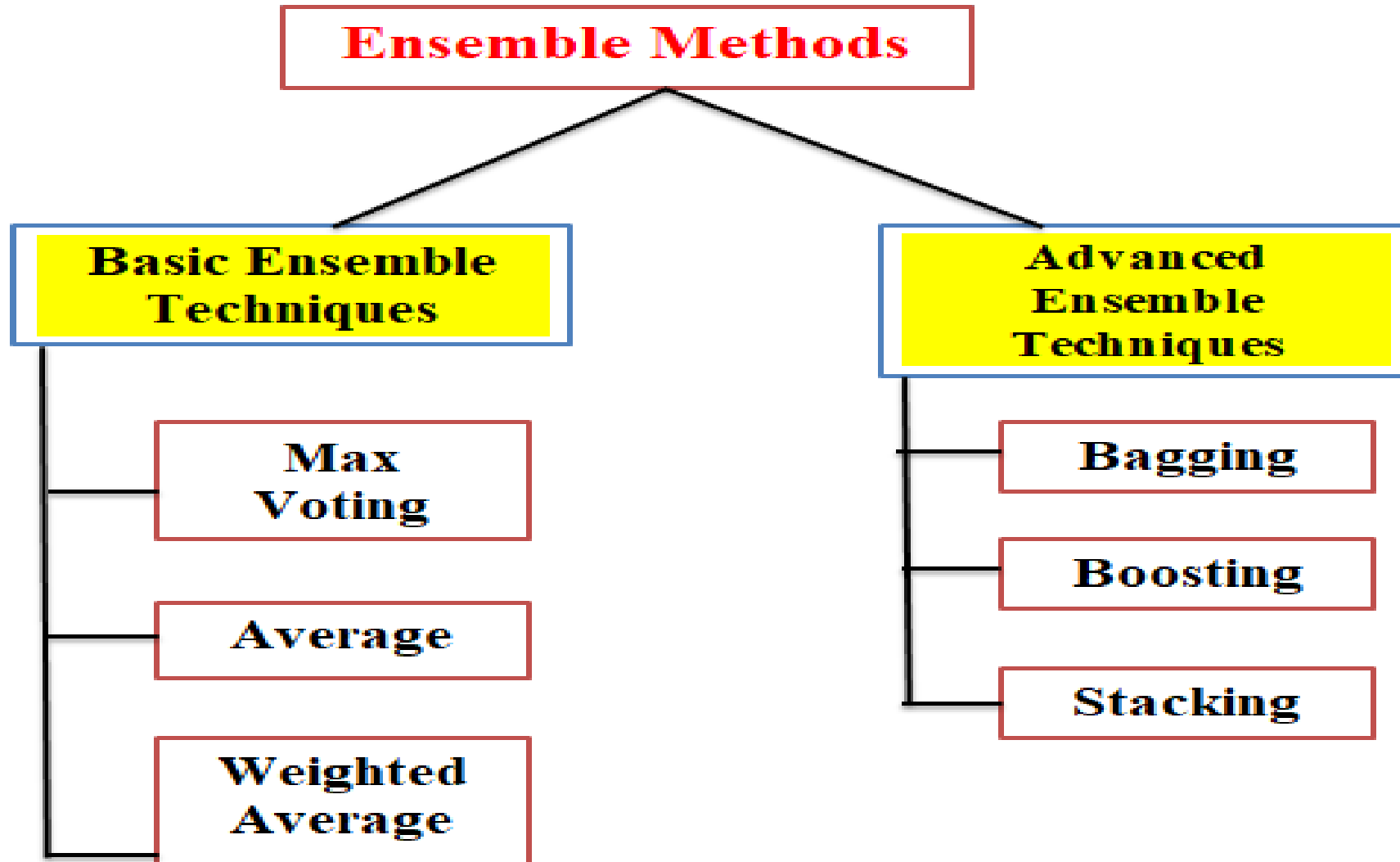


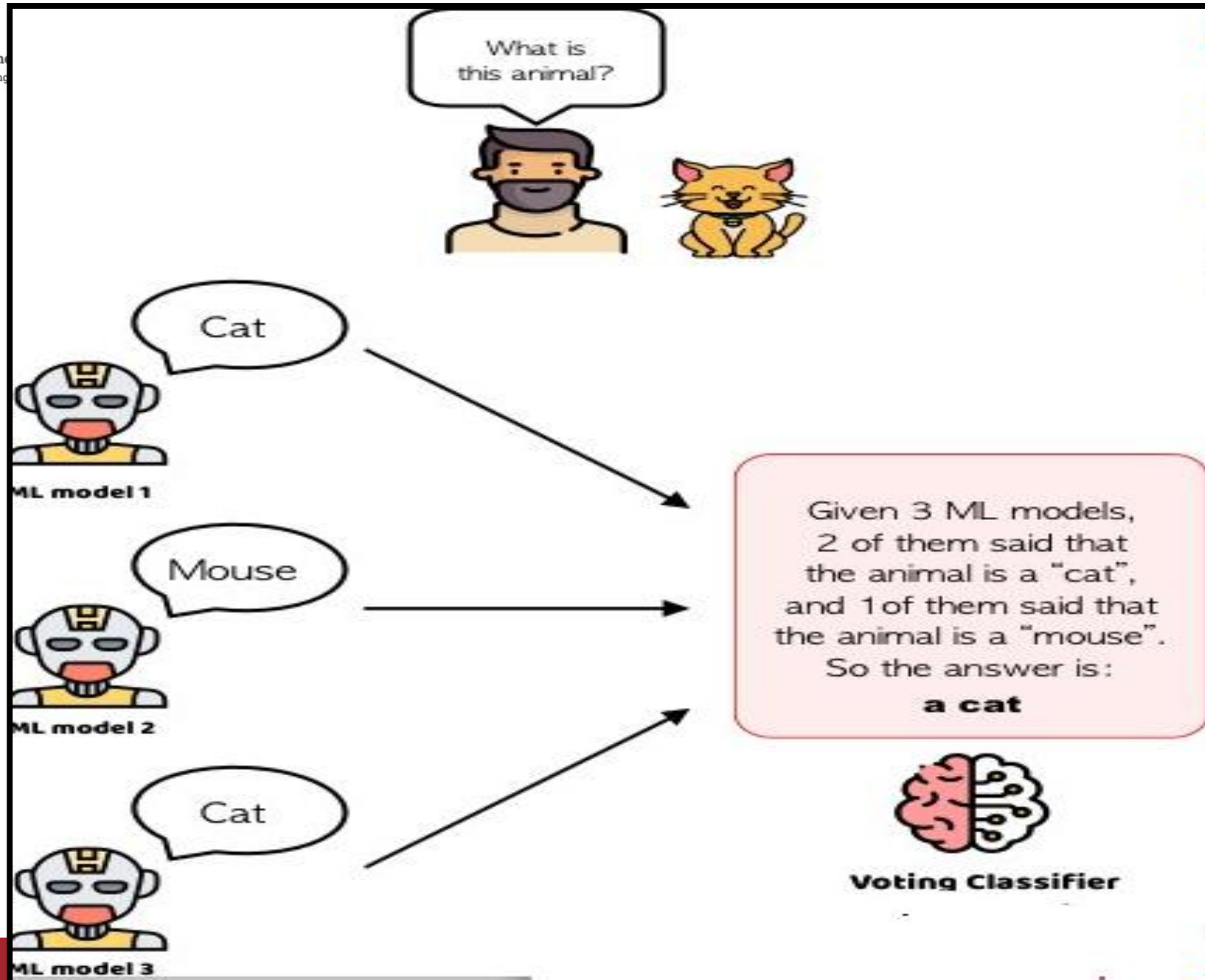
Figure: Types of Ensemble Techniques

- **1. Max Voting:** The **max voting** method is generally used for **classification problems**. In this technique, **multiple models are used to make predictions for each data point**.
- The predictions by each model are considered as **a 'vote'**.
- The predictions which we get from the **majority of the models** are used as the **final prediction**.
- **For Example:** when you **asked 5 of your colleagues** to **rate your movie (out of 5)**;
- we'll assume **three of them rated it as 4** while **two of them** gave **it a 5**.

Contd.

- Since the **majority gave a rating of 4**, the **final rating will be taken as 4**. You can consider this as taking the mode of all the predictions.
- **The result of max voting would be something like this:**

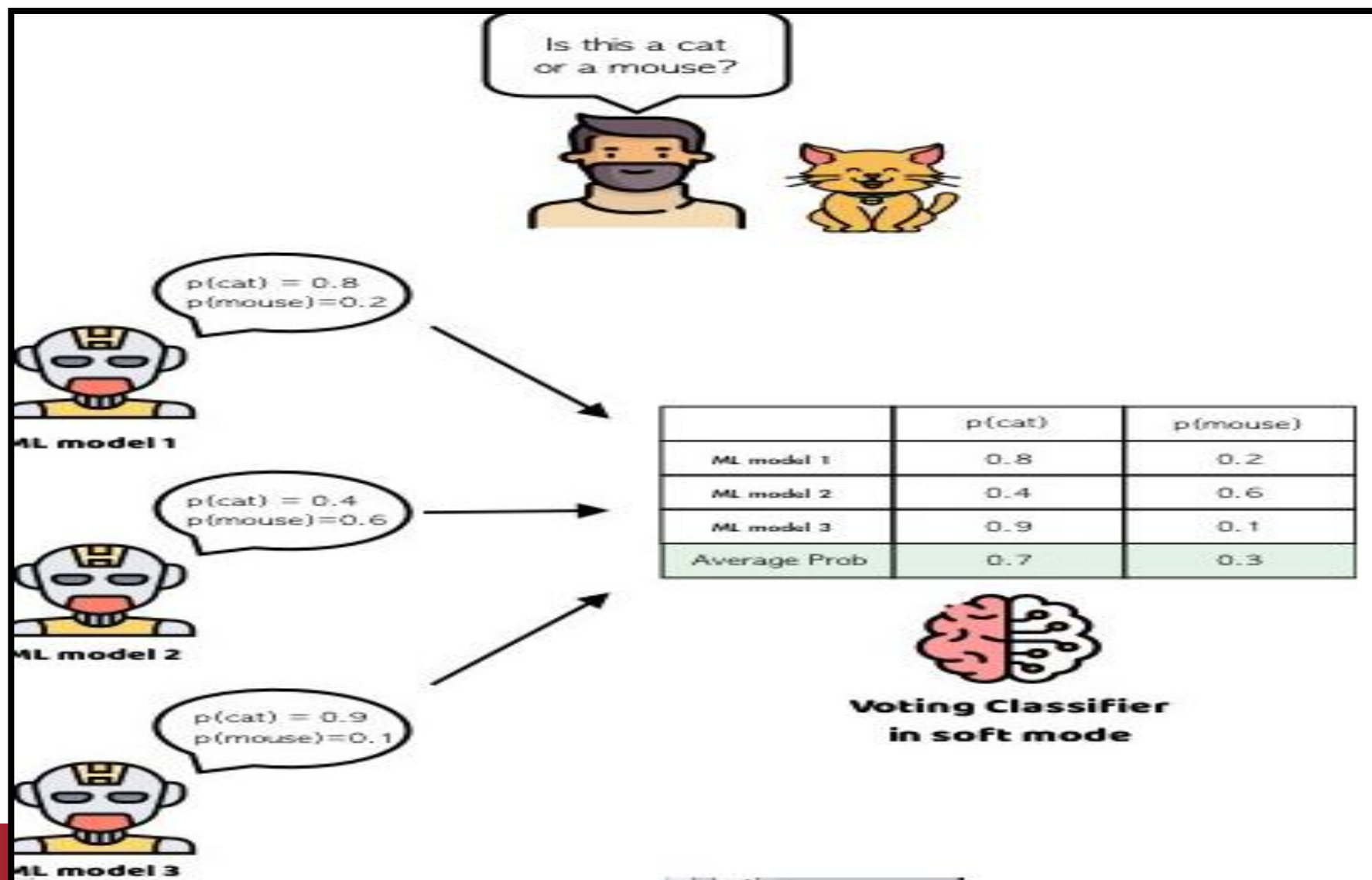
Colleague 1	Colleague 2	Colleague 3	Colleague 4	Colleague 5	Final Rating
5	4	5	4	4	4



- **2. Average Voting:** Similar to the **max voting technique**, **multiple predictions are made for each data point in averaging.**
- In this method, we take **an average of predictions from all the models** and use it to make the **final prediction.**
- Averaging can be used for making predictions in **regression problems** or while **calculating probabilities** for **classification problems.**
- **For example**, in the below case, the averaging method would take the **average of all the values.**

$$\text{i.e. } (5+4+5+4+4)/5 = 4.4$$

Colleague 1	Colleague 2	Colleague 3	Colleague 4	Colleague 5	Final Rating
5	4	5	4	4	4.4



- **3. Weighted Average:** This is an **extension** of the **averaging method**.
- All models are assigned **different weights** defining the **importance of each model for prediction**.
- For instance, if two of your **colleagues are critics**, while others have **no prior experience in this field**, then the answers by these **two friends** are given more importance as compared to the other people.

Colleague 1	Colleague 2	Colleague 3	Colleague 4	Colleague 5	Final Rating	
Weight	0.23	0.23	0.18	0.18	0.18	
Rating	5	4	5	4	4	4.41

➤ The result is calculated as

$$[(5*0.23) + (4*0.23) + (5*0.18) + (4*0.18) + (4*0.18)] = 4.41.$$

Advanced Ensemble Techniques

- Bagging and Boosting are the two very important ensemble methods to improve the measure of accuracy in predictive models which is widely used.
- While performing a machine learning algorithm we might come across various errors such as noise . To overcome these errors we apply ensemble methods.

- **1. Bagging:** Bagging is an **acronym** for Bootstrapped Aggregation. And it is a **simple and very powerful ensemble method**.
- It is mainly used in **Decision Trees**.
- **For Eg:** In the analogy, each friend will take the test separately, and then **their answers will be combined** to **form the final answer on the exam**. There are many ways of combining the answers, but the **most common way is voting**.
- Bagging is mainly used in Random Forest Algorithm.

- In 1996, [Leo Breiman](#) introduced the bagging algorithm, which has three basic steps:
 - 1. Bootstrapping
 - 2. Parallel Training
 - 3. Aggregation
- 1. Bootstrapping: Bootstrapping is a **sampling technique** in which **subsets of observations** from the **original dataset are generated**.
- This means that **each time you select a data point** from the **training dataset**, you are able to **select the same instance multiple times**.

2. Parallel Training: These bootstrap samples are then **trained independently** and **in parallel with each other using base learners**.

3. Aggregation: Finally, depending on the task(i.e. **regression or classification**), an **average or a majority of the predictions** are **taken** to compute the **Accurate Results**.

- In the **case of regression**, an **average** is taken of all the outputs predicted by the **individual classifiers**; this is known as **soft voting**.
- For **classification problems**, the class with the **highest majority of votes** is **accepted**; this is known as **hard voting or majority voting**.

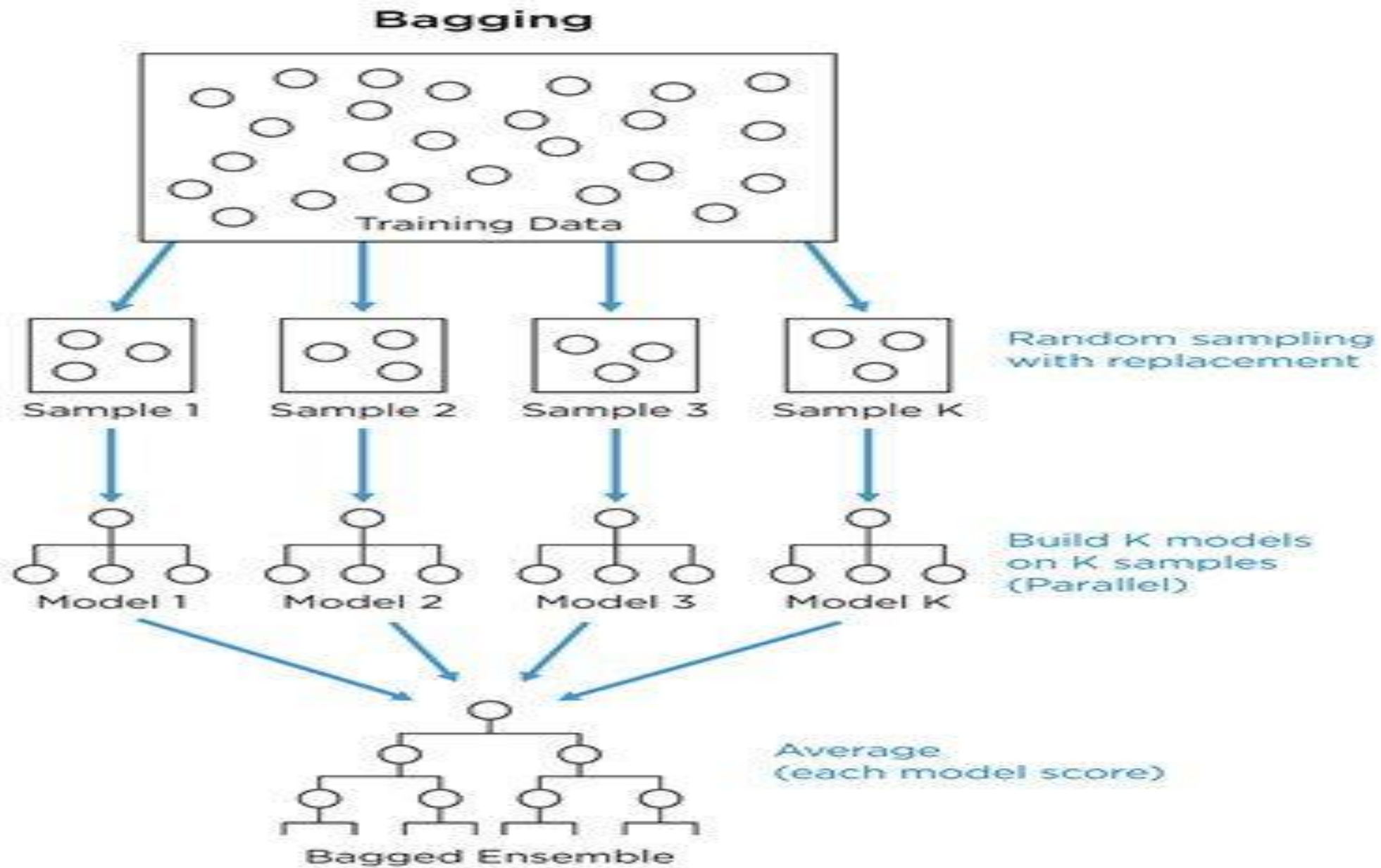
Implementation Steps of Bagging

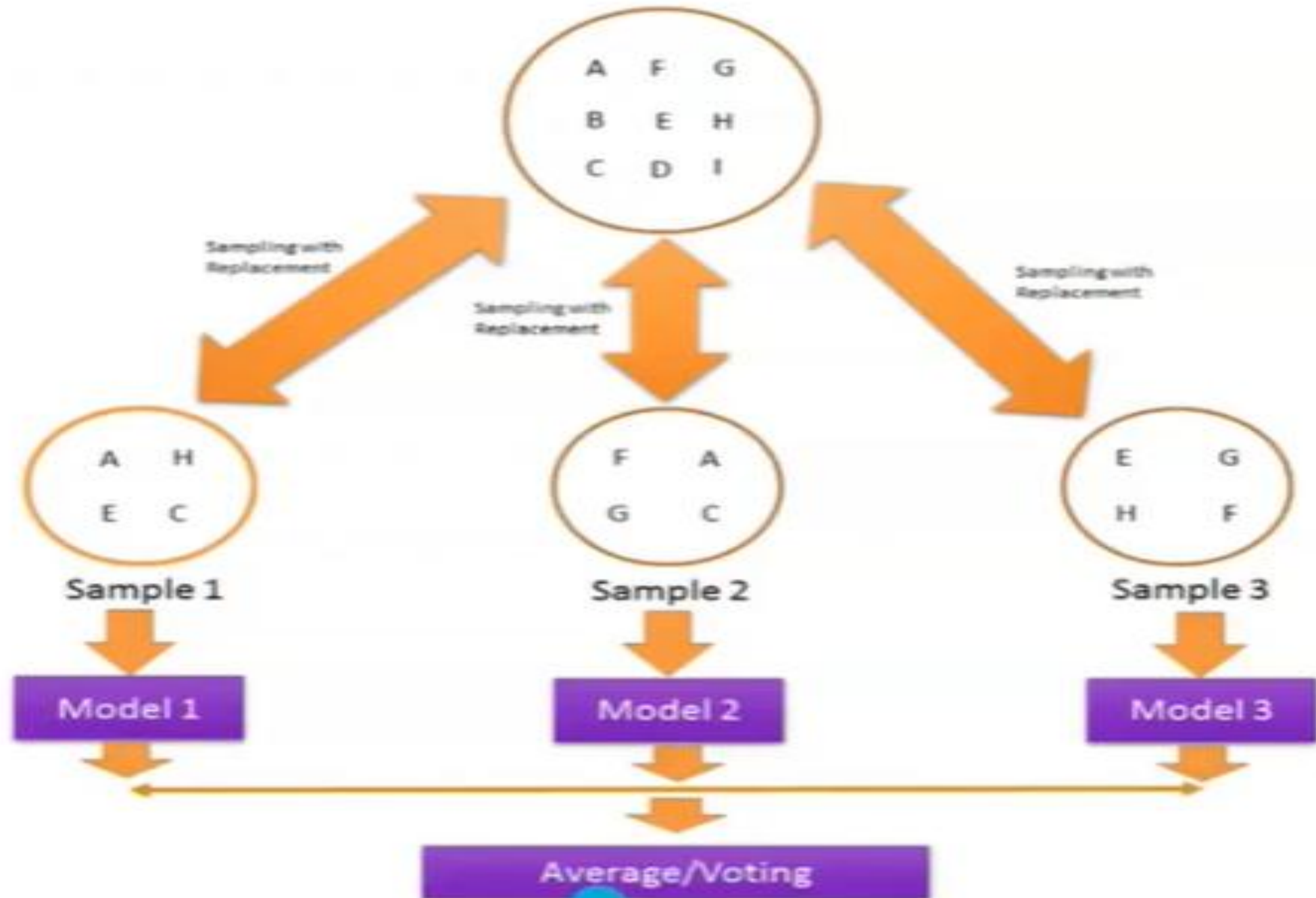
Step 1: Multiple subsets are created from the original data set with equal tuples, selecting observations with replacement.

Step 2: A base model is created on each of these subsets.

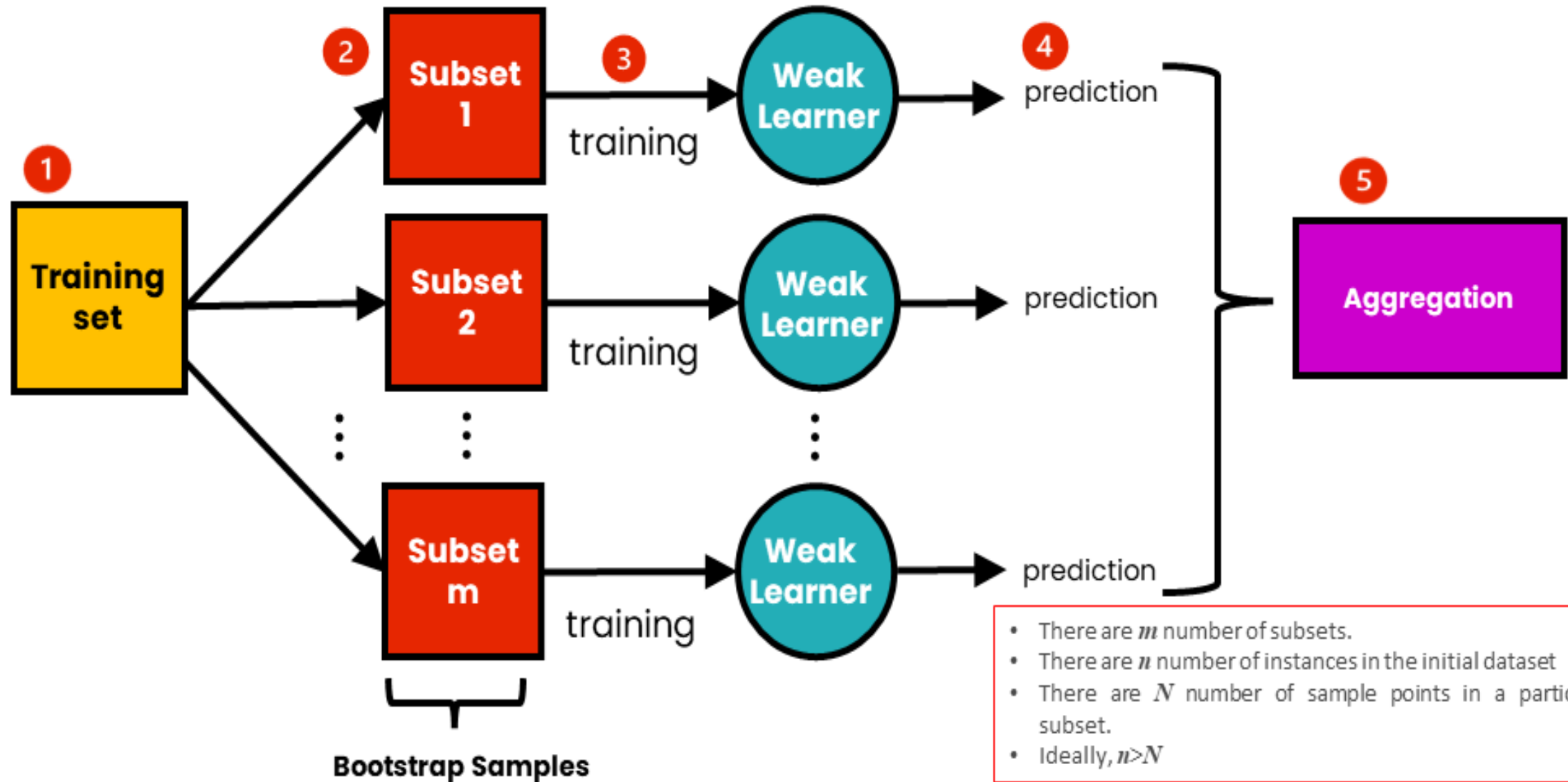
Step 3: Each model is learned in parallel with each training set and independent of each other.

Step 4: The final predictions are determined by combining the predictions from all the models.





The Process of Bagging (Bootstrap Aggregation)



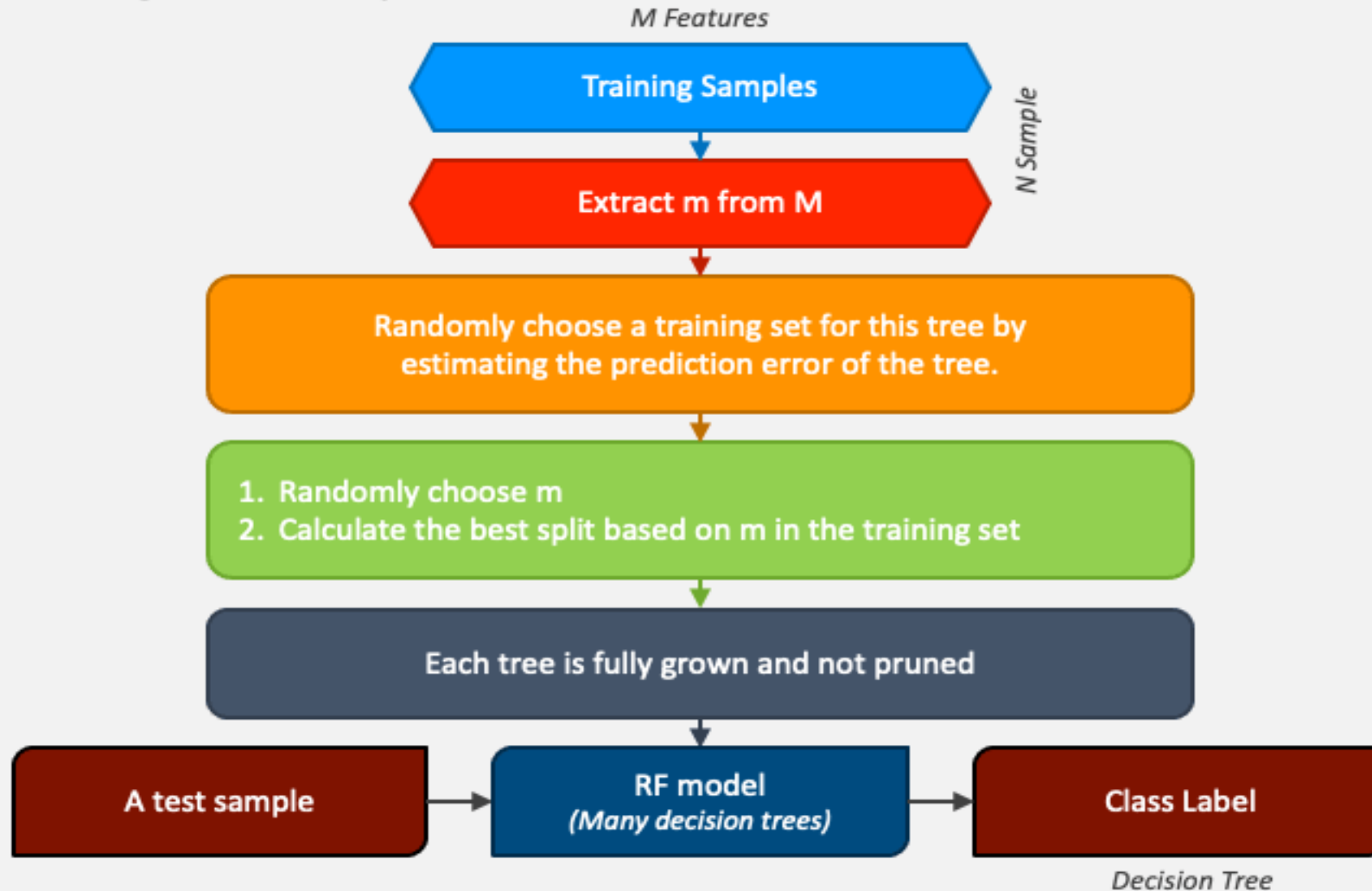
Random Forest Algorithm

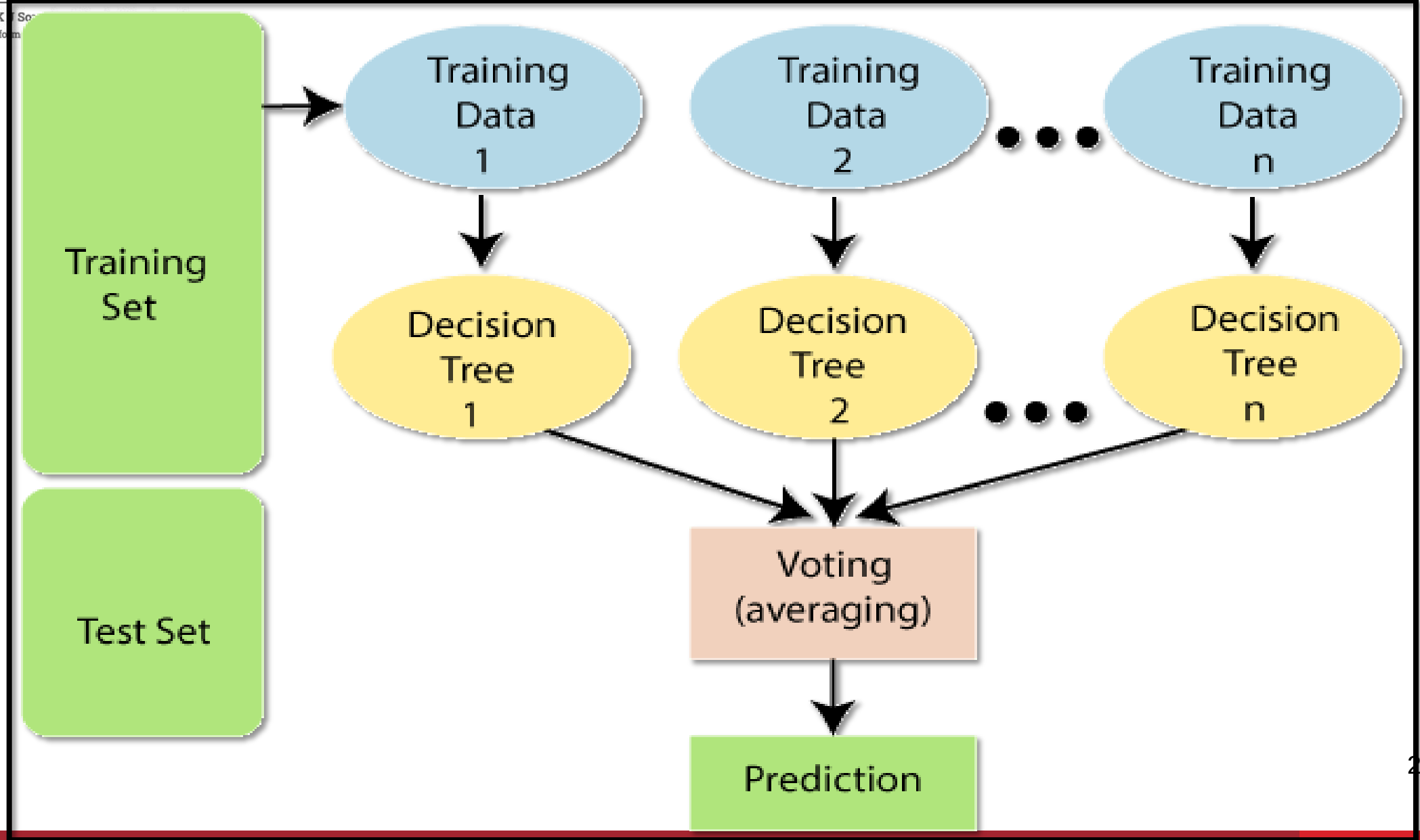
- **Random Forest** is a popular **machine learning algorithm** that belongs to the **supervised learning technique**.
- It can be used for both **Classification and Regression** problems in ML.
- It is based on the concept of **ensemble learning**, which is a process of **combining multiple classifiers** to solve a **complex problem and to improve the performance of the model**.

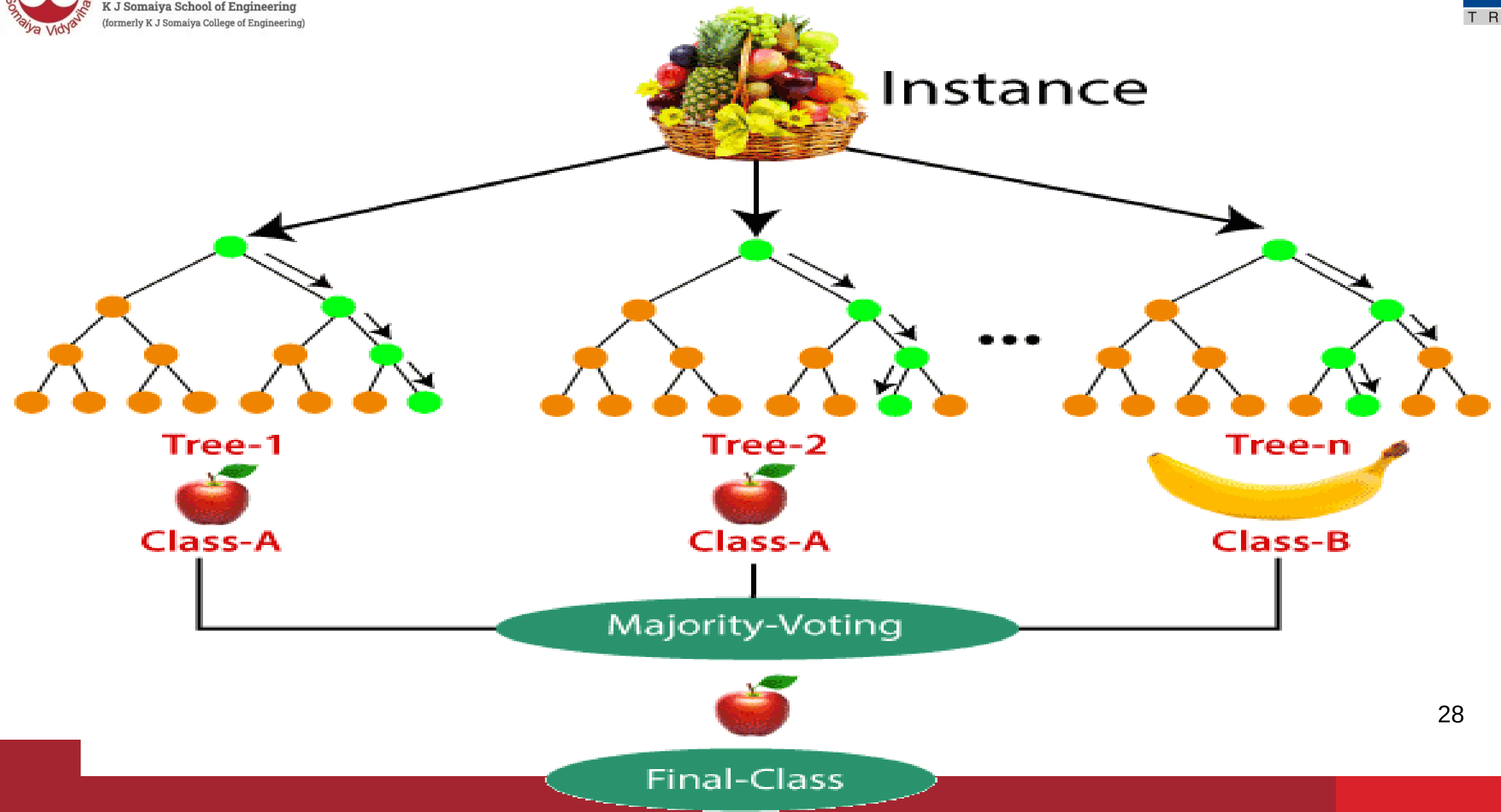
- As the name suggests, "**Random Forest**" is a classifier that contains a **number of decision trees** on various subsets of the given dataset and **takes the average to improve the predictive accuracy of that dataset.**
- Here the random forest takes **the prediction from each tree and based on the majority votes of predictions**, and it **predicts the final output.**
- The **main Aim is** the **greater number of trees in the forest** leads to higher accuracy and **prevents the problem of overfitting.**

RANDOM FOREST

Random Forest: Algorithm Steps







There are mainly **four sectors** where **Random forest mostly used:**

- 1. Banking:** Banking sector mostly uses this algorithm for the **identification of loan risk.**
- 2. Medicine:** With the help of this algorithm, **disease trends and risks of the disease can be identified.**
- 3. Land Use:** We can identify the **areas of similar land use by this algorithm.**
- 4. Marketing:** **Marketing trends** can be identified using this algorithm.

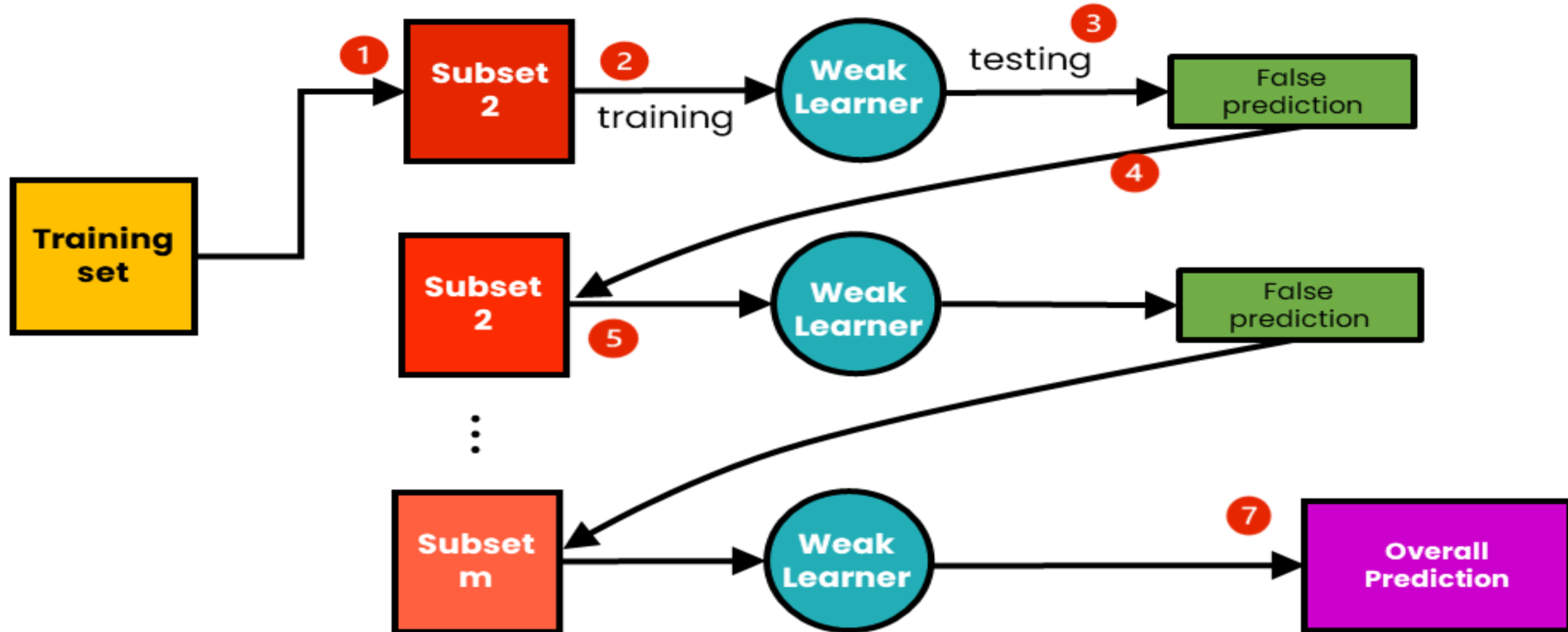
- **2.Boosting:** Gradient Boosting is an **ensemble learning technique** in which **models are not run independently** but **sequentially**.
- Boosting is a kind of algorithm that is able to **convert weak learners to strong learners**.
- In Boosting Technique, Each Algorithm. Ie, **base learners are trained sequentially and at every time the next learner is trying to reduce the error** by **updating the parameters** and perform better comparison to the previous learners.

How Boosting Works

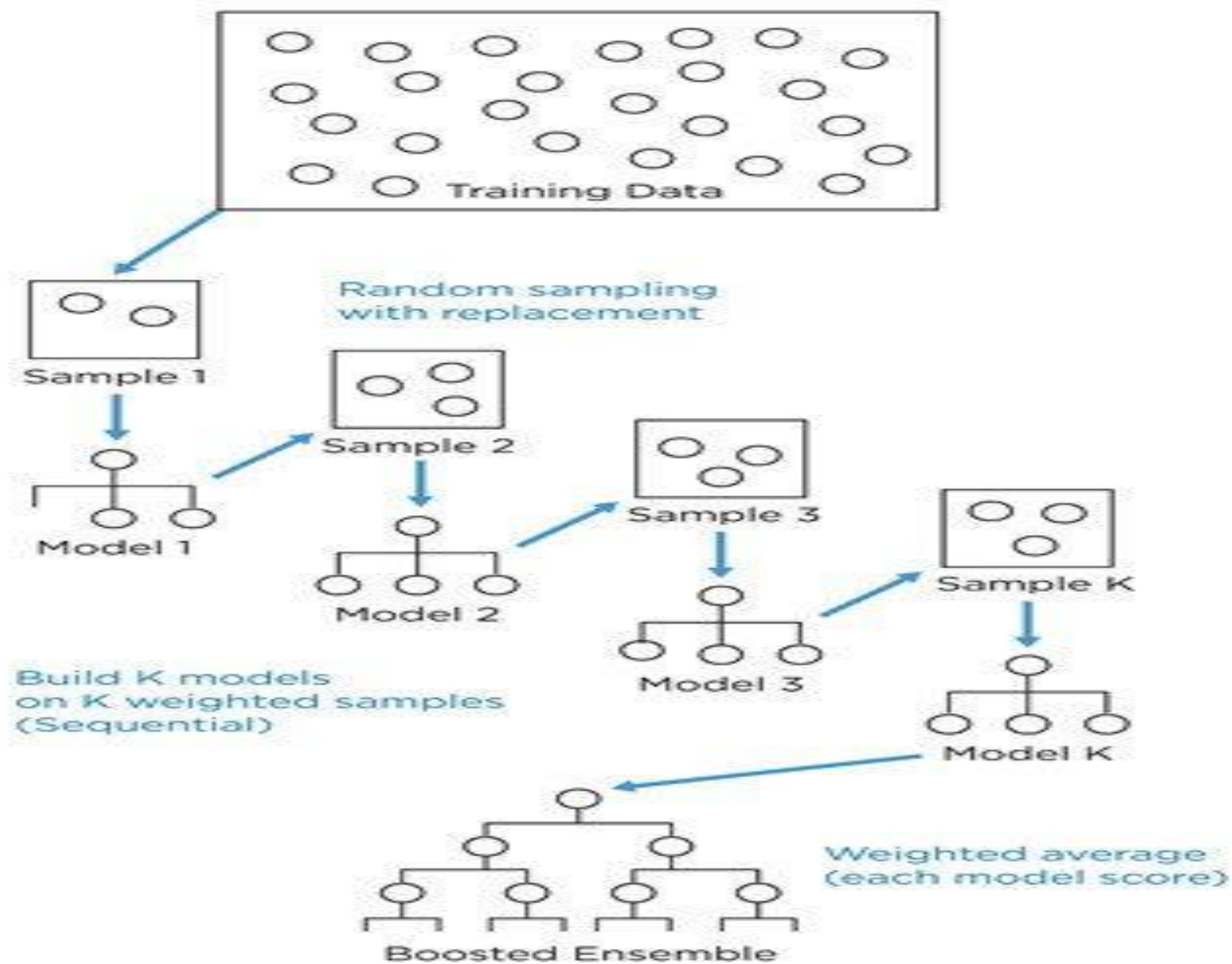
- Here's what the entire process looks like:
 - 1. Create a subset from the original data,
 - 2. Build an initial model with this data,
 - 3. Run predictions on the whole data set,
 - 4. Calculate the error using the predictions and the actual values,
Assign more weight to the incorrect predictions.

- Create another model that attempts **to fix errors from the last model**,
- Run predictions on the **entire dataset with the new model**,
- Create several models with each model aiming at correcting the errors generated by the previous one, Obtain the **final model by weighting the mean of all the models**.
- Boosting algorithms is the family of algorithms that combine weak learners into a strong learner.

The Process of Boosting



Boosting

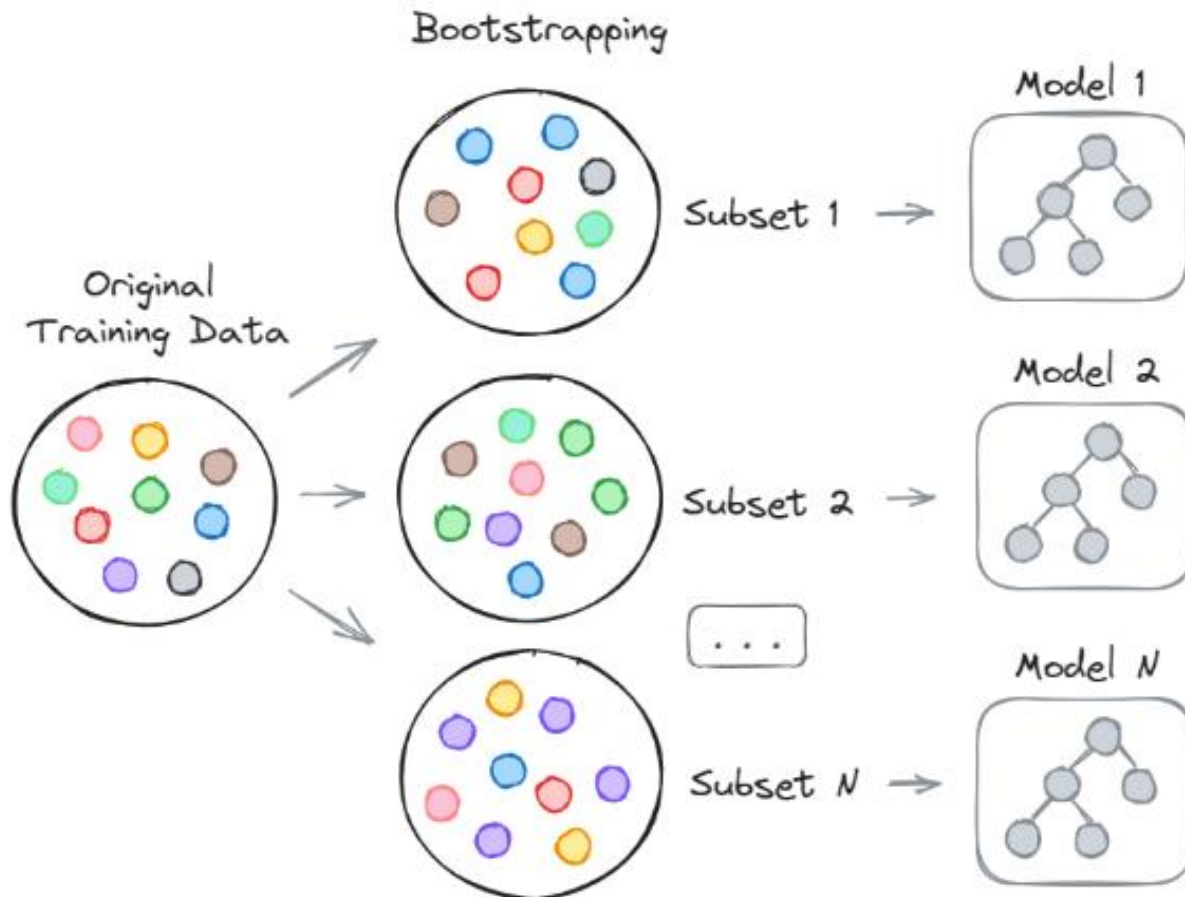


Differences between Bagging And Boosting

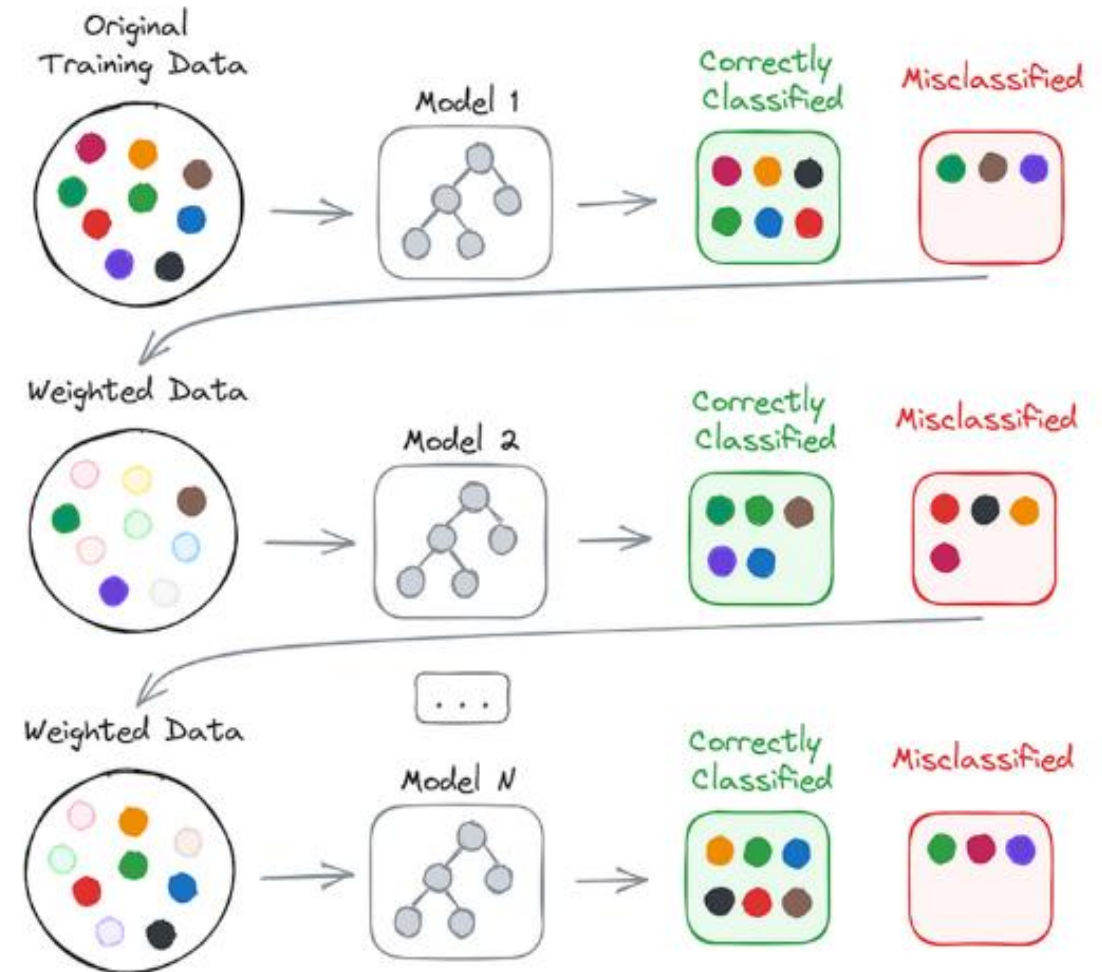
	Bagging	Boosting
Similarities	• Uses voting • Combines models of the same type	
Differences	Individual models are built separately	Each new model is influenced by the performance of those built previously
	Equal weight is given to all models	Weights a model's contribution by its performance

Differences between Bagging And Boosting

Bagging

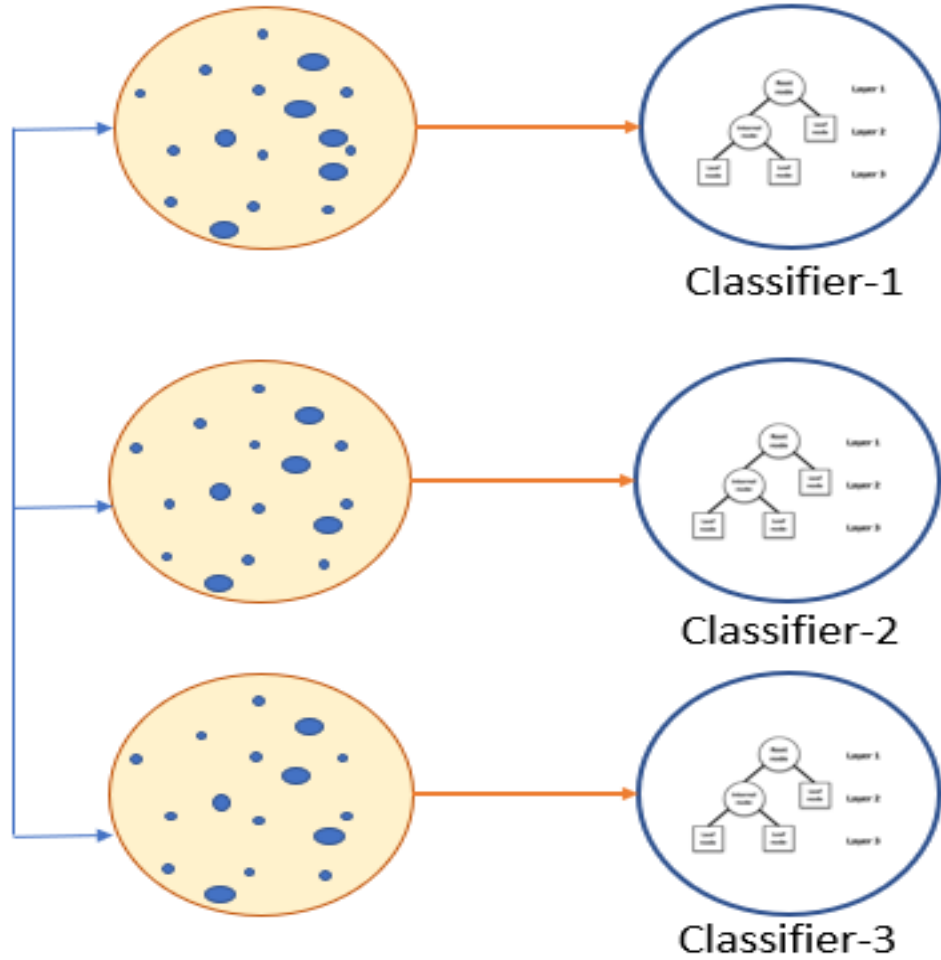


Boosting



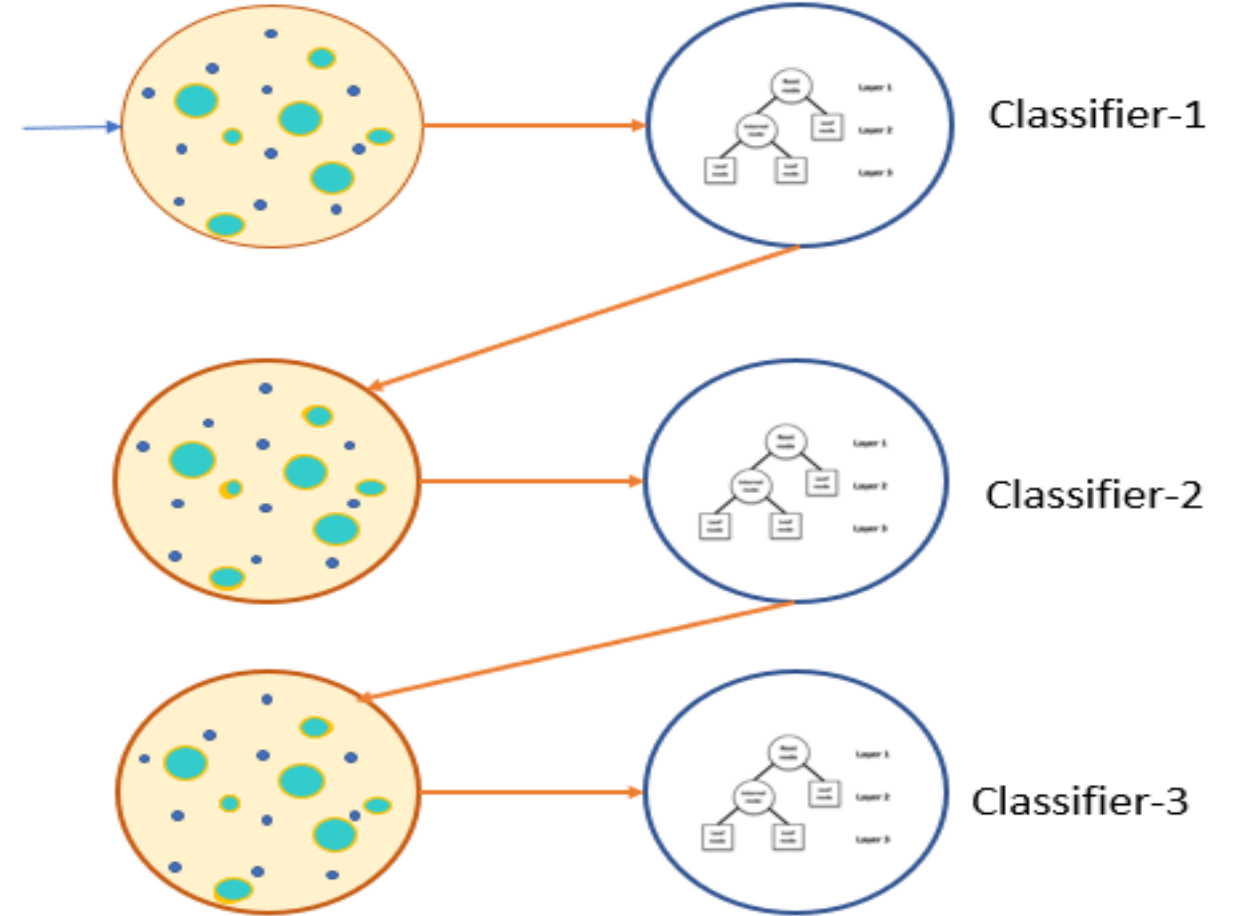
Differences between Bagging And Boosting

Bagging



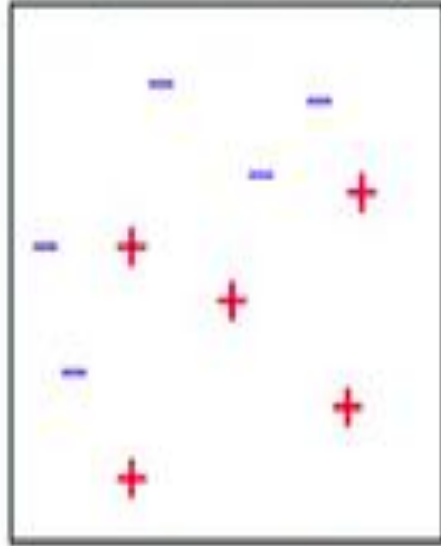
Parallel

Boosting

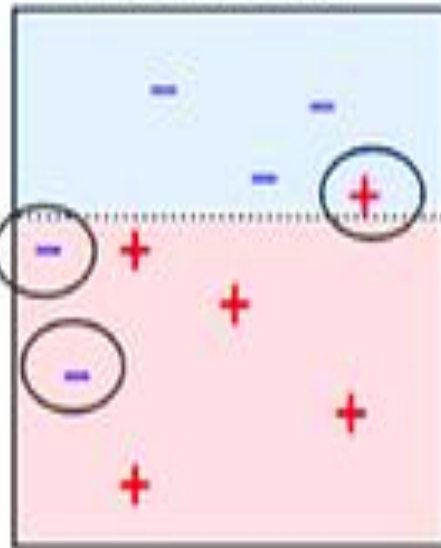


Sequential

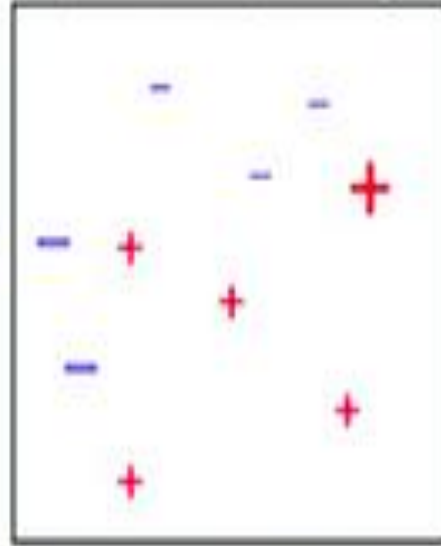
Original data set, D_1



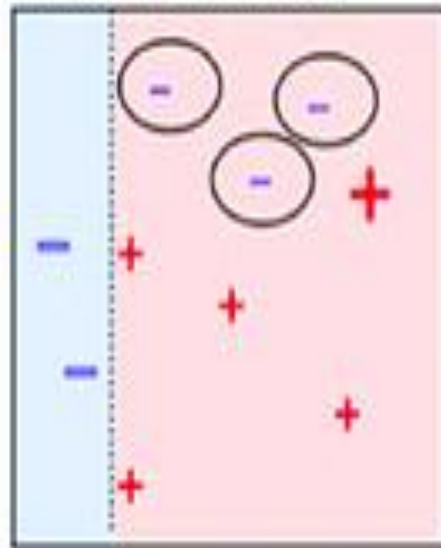
Trained classifier



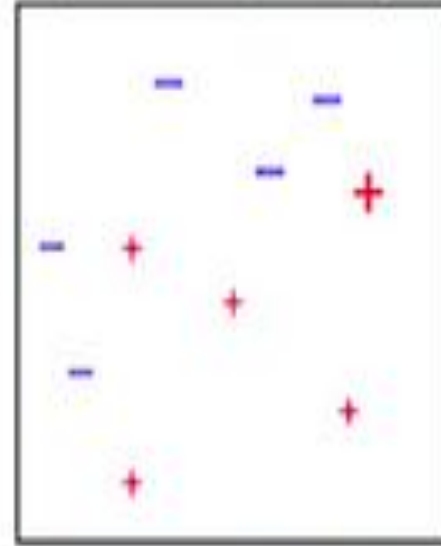
Update weights, D_2



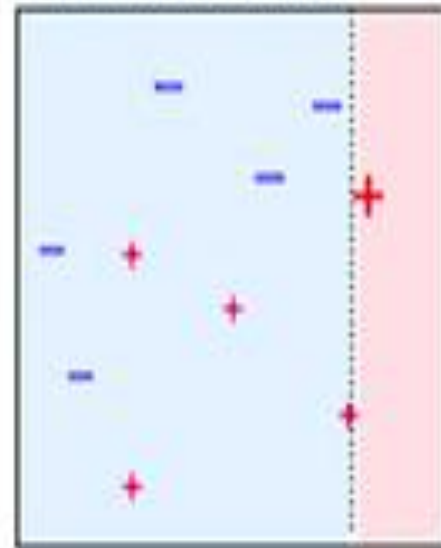
Trained classifier



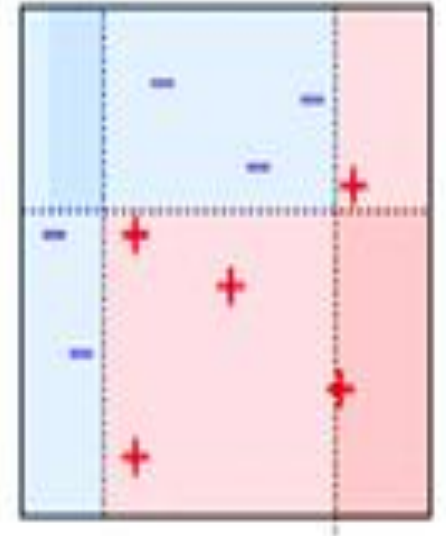
Update weights, D_3



Trained classifier



Combined classifier



Main Steps involved in boosting are :

- **Train model A** on the **whole set**
- **Train the model B** with exaggerated data on the **regions in which A performs poorly**
- ...
- Instead of **training the models in parallel**, we can **train them sequentially**. This is the **main idea of Boosting!**

- In order to provide you with an **idea about mail spam detection problem.**
- The **spam detection problem** can be divided into the following steps.
 - If the email **contains only an image?**
 - **Who is the sender?**
 - How **caps lock** was used in the email?
 - Check the **subject line in the email**